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(54) **SYSTEMS AND METHODS FOR
LOCATION-BINDING AUTHENTICATION**

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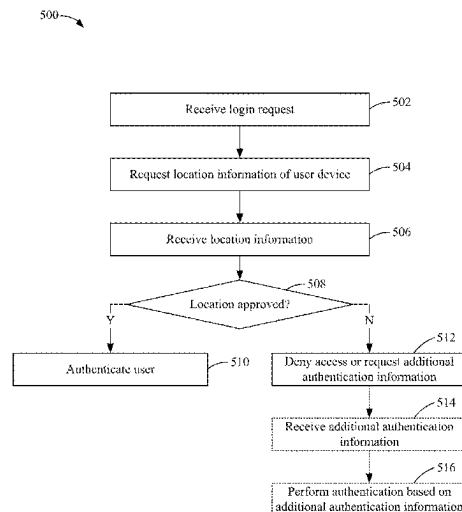
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(57) **ABSTRACT**

Systems, methods, and apparatuses for authenticating a user based at least in part on a location of the user or a location of a user device are described. A method includes: receiving a login request including a user identifier associated with a user; transmitting a request for location information of a user device associated with the user; causing computer-executable code deployed to the user device to generate a location-based modifiable digital fingerprint comprising a first encoded value based on a unique identifier of the user device and a second encoded value based on a first location information of the user device; receiving the location-based modifiable digital fingerprint from the user device; verifying that the location information from the location-based modifiable digital fingerprint corresponds to a second location; and providing the user access to the computing device.

20 Claims, 5 Drawing Sheets



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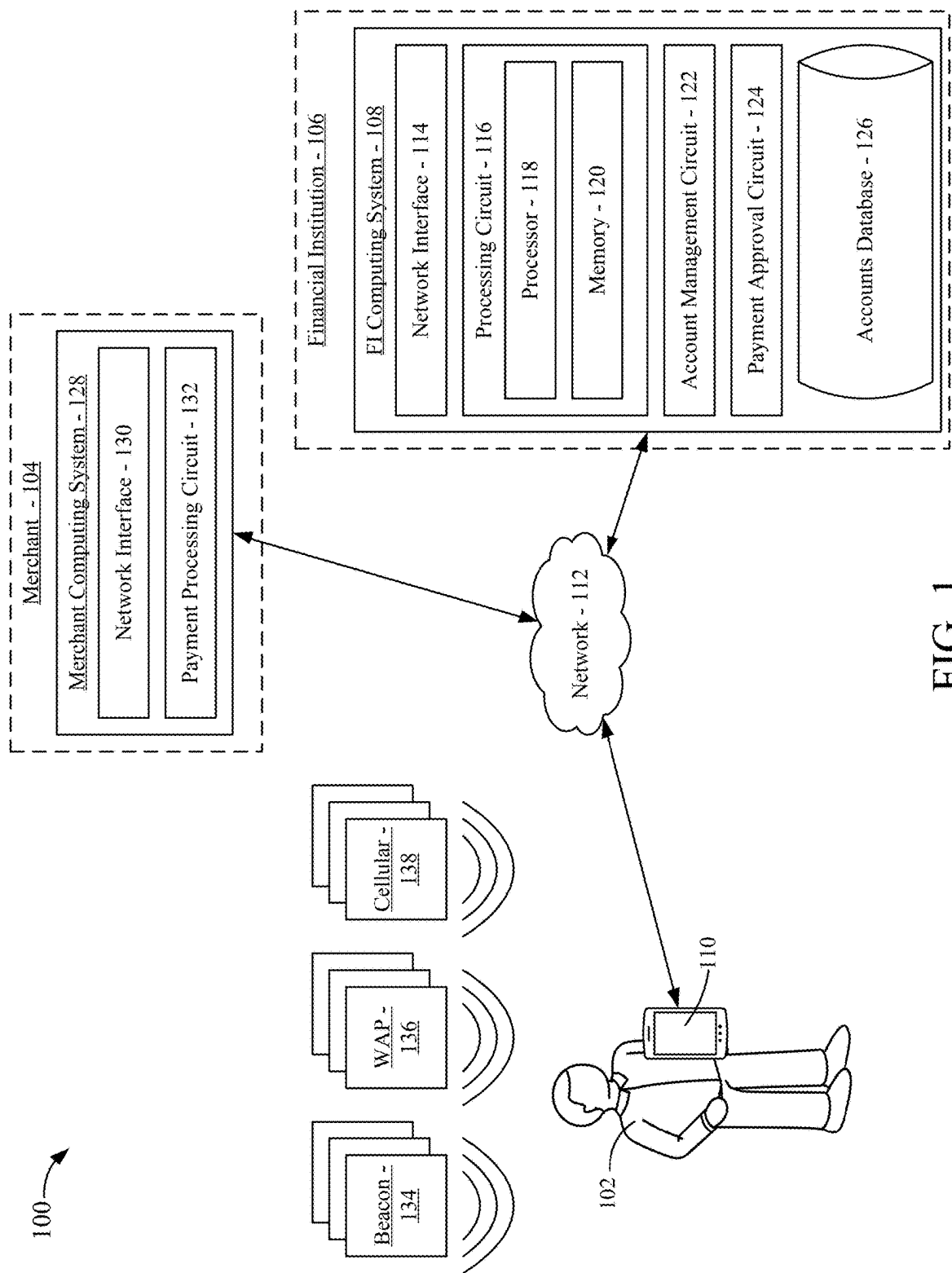


FIG. 1

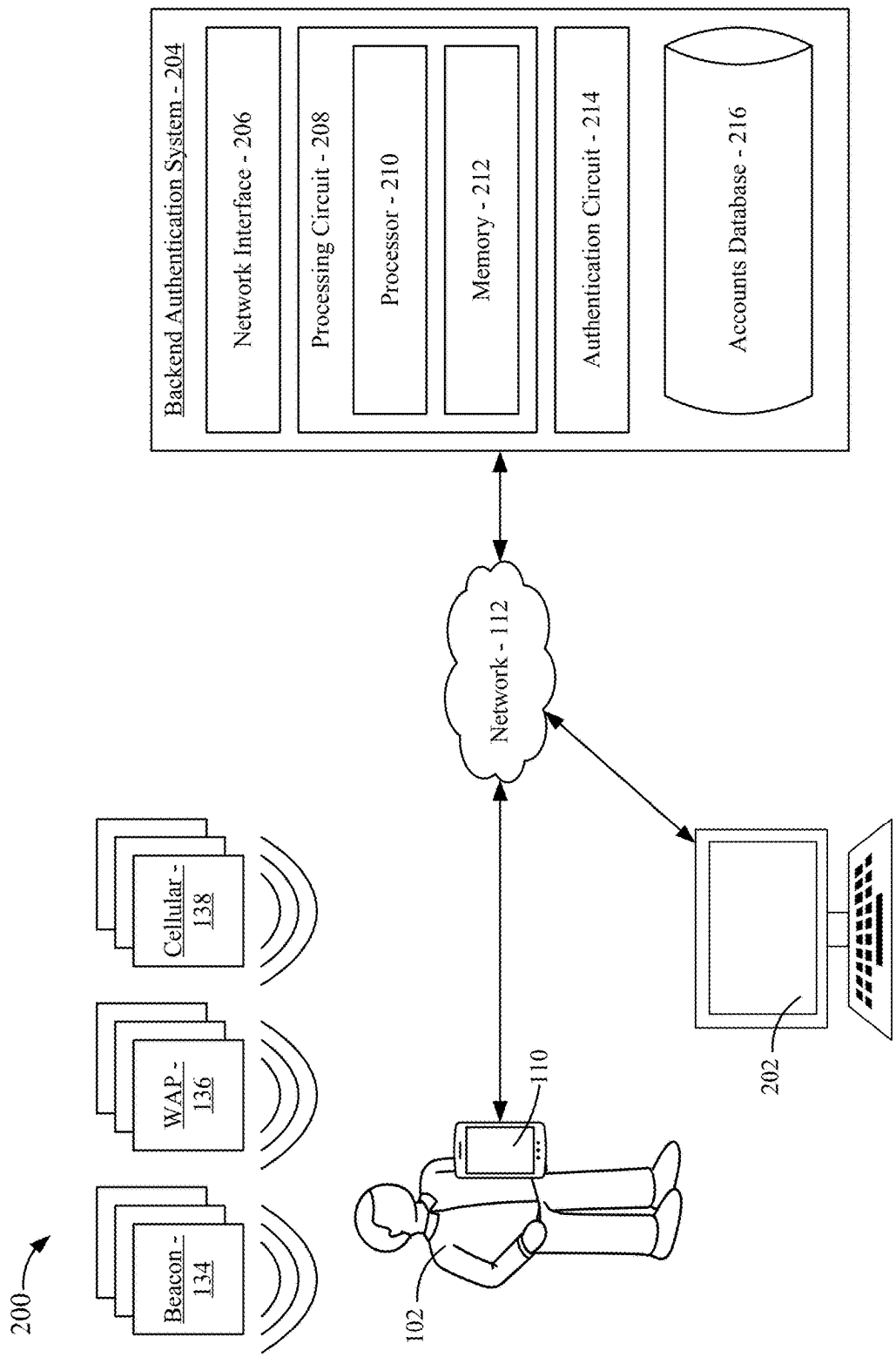


FIG. 2

300

302	Device ID:	1X478VH390184JJ3
304	Time:	MARCH 3, 2016 15:34 EST
306	<u>LOCATION SNAPSHOT</u>	
308	GPS Sat	Distance
	9	#####
	30	#####
	23	#####
	28	#####
310	WiFi SSID	Strength
	Coffee Shop	-64.3db
	Work-Private	-68.4db
	Work-Guest	-68.4db
	Apartment12	-61.2db
312	Bluetooth Beacon ID	
	132DF5432H12323	
314	Other	
316	QR Code	12345WORKPLACE
	NFC Tag ID	N/A

FIG. 3

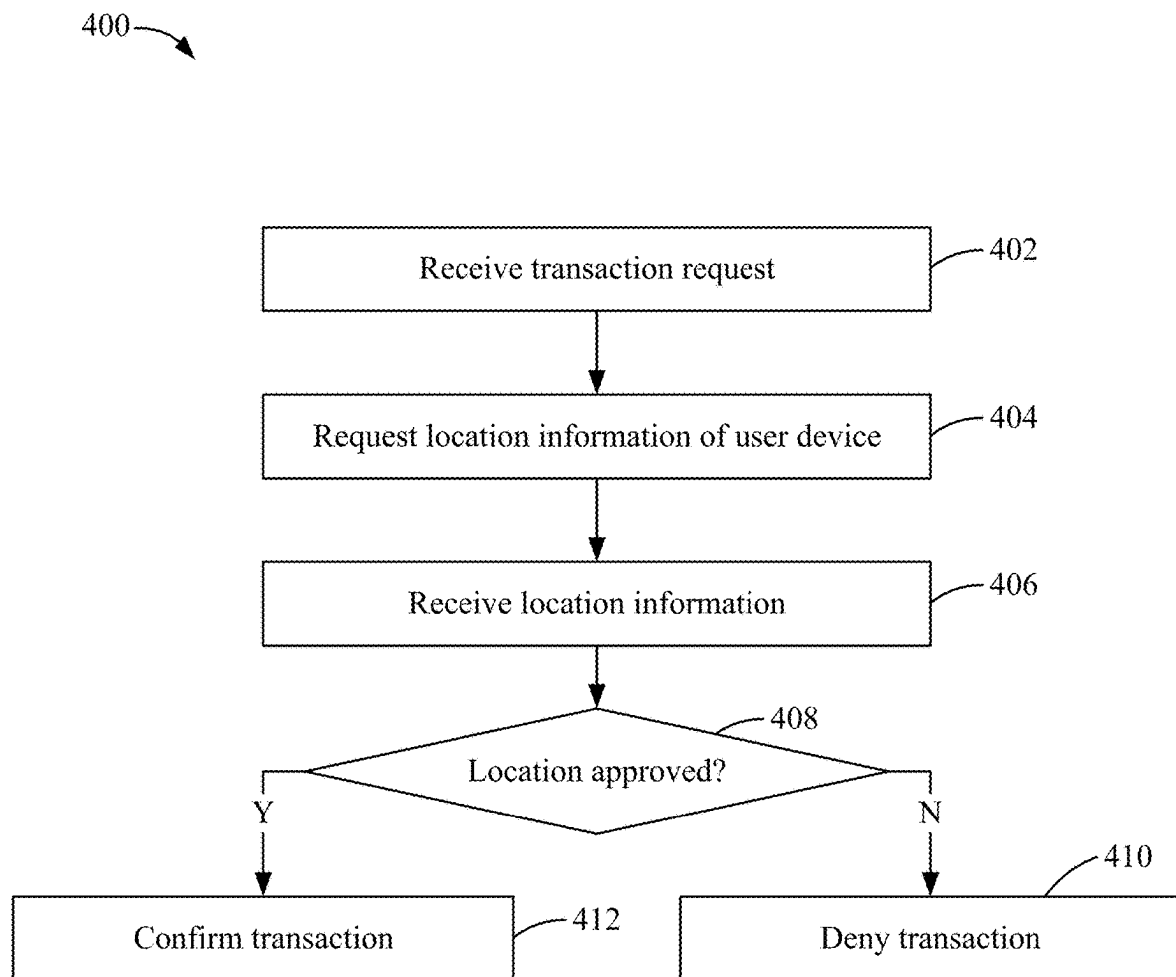


FIG. 4

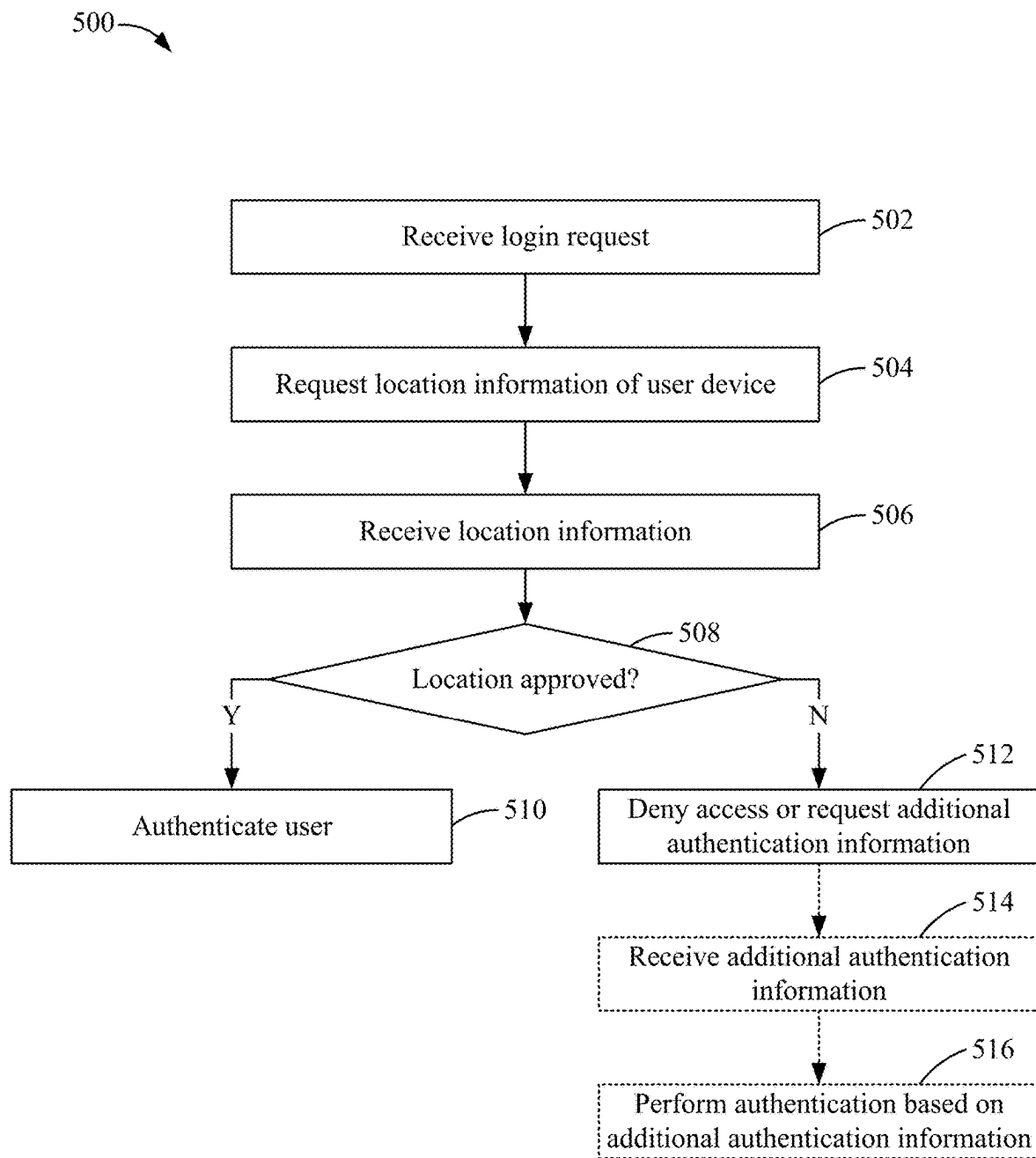


FIG. 5

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SYSTEMS AND METHODS FOR LOCATION-BINDING AUTHENTICATION

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 15/204,649, filed Jul. 7, 2016, now U.S. Pat. No. 11,132,425, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

Embodiments of the present disclosure relate to systems and methods for using location-based authentication.

BACKGROUND

Many systems require users to authenticate themselves prior to gaining access to the system. For example, many financial institutions provide online services to their customers via online banking portals and mobile banking applications that allow the customers to remotely manage their financial accounts and complete various financial transactions through internet after the customers are fully authenticated. Additionally, many employers require that their employees authenticate themselves via employee work terminals (e.g., computers, laptops, etc.) prior to providing access to the underlying system. The user authentication is required to protect user privacy and to reduce fraud. Many authentication processes may require inputs of authentication codes, such as passwords, PINs, authentication codes (e.g., dynamically generated security token), biometrics, security question answers, etc.

SUMMARY

A first example embodiment relates to a method of authenticating a login request at a computing device. The method includes receiving, at an authentication system associated with the computing device, a login request from the computing device. The login request includes a user identifier associated with the user. The method further includes transmitting, by the authentication system, a request for location information of a user device associated with the user. The method includes receiving, by the authentication system, location information from the user device. The method further includes verifying, by the authentication system, that the location information received from the user device corresponds to a known location of the computing device. The method includes providing, by the authentication system, the user access to the computing device.

Another example embodiment relates to a system. The system includes a computing device and a backend authentication system in communication with the computing device via a network. The backend authentication system includes a processing circuit having a processor and memory. The processing circuit is structured to receive a login request from the computing device. The login request includes a user identifier associated with a user attempting to gain access to the computing device. The processing circuit is further structured to transmit a request for location information to a user device associated with the user. The processing circuit is structured to receive location information from the user device. The processing circuit is further structured to verify that the location information received

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from the user device corresponds to a known location of the computing device and to provide the user access to the computing device.

A further example embodiment relates to a method of authenticating a transaction request involving an originator and a recipient. The method includes receiving, by a financial institution computing system associated with a financial institution where the originator maintains an account involved in the transaction, a transaction request. The method further includes transmitting, by the financial institution computing system, a request for location information to a user device associated with the originator. The method includes receiving, by the financial institution computing system, location information from the user device. The method further includes verifying, by the financial institution computing system, that the location information received from the user device corresponds to an approved location for the transaction. The method includes approving, by the financial institution computing system, the transaction.

Another example embodiment relates to a financial institution computing system associated with a financial institution. The system includes a network interface structured to facilitate data communication via a network. The system further includes an accounts database structured to store information associated with accounts held by the financial institution. The system includes a processing circuit comprising a processor and memory. The processing circuit is structured to receive a transaction request relating to a transaction involving a transfer of funds from an originator to a recipient, the originator having an account with the financial institution. The processing circuit is further structured to transmit a request for location information to a user device associated with the originator. The processing circuit is structured to receive location information from the user device. The processing circuit is further structured to verify that the location information received from the user device corresponds to an approved location for the transaction and to approve the transaction.

These and other features, together with the organization and manner of operation thereof, will become apparent from the following detailed description when taken in conjunction with the accompanying drawings.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 is a view of a system that facilitates a transfer of funds from a user to a recipient according to an example embodiment.

FIG. 2 is a view of a user authentication system according to an example embodiment.

FIG. 3 shows a table of information used to form a digital fingerprint of a user device of the system of FIG. 1 or the system of FIG. 2.

FIG. 4 is a flow diagram of a method of authenticating a financial transaction according to an example embodiment.

FIG. 5 is a flow diagram of a method of authenticating a user according to an example embodiment.

DETAILED DESCRIPTION

Referring to the figures generally, systems, methods, and apparatuses for authenticating a user based at least in part on a location of the user or a location of a user device are described herein. The user may be authenticated as part of a financial transaction or as part of a login process for a computing device. The user device was previously bound or

associated with the user. During the authentication, a system (e.g., a financial institution computing system or a backend authentication system) requests location information from the user device. The location information may be packaged as a digital fingerprint of the user device, which can only be created from the user device. Based on the location information, the user can be authenticated thereby approving the transaction or login request.

Referring to FIG. 1, a view of a system **100** is shown according to an example embodiment. As described below in further detail, the system **100** facilitates a transfer of funds from a user **102** to a recipient. In the specific arrangement of FIG. 1, the recipient is a merchant **104**, and the transfer of funds is described within the context of a purchase by the user **102** from the merchant **104** (e.g., an online purchase). The transfer of funds is facilitated by a financial institution **106**. It should be understood that the merchant **104** may be replaced with any type of entity that receives funds from the user **102** (e.g., a financial institution that receives funds from the user **102** in a transfer between financial institutions not necessarily in connection with the purchase of goods or services).

The user **102** is an account holder with the financial institution **106**. The financial institution **106** includes a financial institution (FI) computing system **108**. The FI computing system **108** maintains information about accounts held with the financial institution **106** and facilitates the movement of funds into and out of the accounts. The user **102** can manage and maintain (e.g., view balances, initiate transfers, change contact information, etc.) the account with the financial institution **106** via the user device **110**. The user device **110** may be a mobile device (e.g., a smartphone, a tablet computer), a laptop computer, a desktop computer, or the like. Accordingly, the user **102** can access the financial institution computing system **108** through a website associated with the financial institution **106** (e.g., via a web browser being executed on the user device **110**) or a banking application offered by the financial institution **106** and being executed on the user device **110**. The user device **110** communicates with the FI computing system **108** via a network **112**. In some arrangements, the network **112** includes the Internet.

The financial institution computing system **108** includes a network interface **114**. The network interface **114** is structured to facilitate data communication with other computing systems (e.g., the user device **110**) via the network **112**. The network interface **114** includes hardware and program logic that facilitates connection of the FI computing system **108** to the network **112**. For example, the network interface **114** may include a wireless network transceiver (e.g., a cellular modem, a Bluetooth transceiver, a WiFi transceiver, etc.) and/or a wired network transceiver (e.g., an Ethernet transceiver). In some arrangements, the network interface **114** includes the hardware and programming logic sufficient to support communication over multiple channels of data communication (e.g., the Internet and an internal financial institution network). Further, in some arrangements, the network interface **114** is structured to encrypt data sent over the network **112** and decrypt received encrypted data.

The financial institution computing system **108** includes a processing circuit **116** having a processor **118** and memory **120**. The processor **118** may be implemented as a general-purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a digital signal processor (DSP), a group of processing components, or other suitable electronic processing components. The memory **120** includes one or more

memory devices (e.g., RAM, NVRAM, ROM, Flash Memory, hard disk storage, etc.) that store data and/or computer code for facilitating the various processes described herein. Moreover, the memory **120** may be or include tangible, non-transient volatile memory or non-volatile memory.

The FI computing system **106** includes an account management circuit **122** and a payment approval circuit **124**. Although shown as separate circuits in FIG. 1, in some arrangements, the account management circuit **122** and/or the payment approval circuit **124** are part of the processing circuit **116**. Other arrangements may include more or less circuits without departing from the spirit and scope of the present disclosure. Further, some arrangements may combine the activities of one circuit with another circuit to form a single circuit. Therefore, those of ordinary skill in the art will appreciate that the present arrangement is not meant to be limiting. The account management circuit **122** is structured to perform various account management functions, including maintaining an accounts database **118**, updating account balances, applying interest to accounts, processing payments related to accounts, and the like. The payment approval circuit **124** is structured to approve payment requests relating transfers of funds out of accounts associated with customers (e.g., the user **102**).

The FI computing system **106** includes the accounts database **126**. In some arrangements, the accounts database **126** is part of the memory **120**. The accounts database **126** is structured to hold, store, categorize, and otherwise serve as a repository for information associated with accounts (e.g., loan accounts, savings accounts, checking accounts, credit accounts, etc.) held by the financial institution **106**. For example, the accounts database **126** may store account numbers, account balances, account ownership information, and the like. The accounts database **126** is structured to selectively provide access to information relating to accounts at the financial institution **106** (e.g., to the user **102** via the user device **110**).

Still referring to FIG. 1, the merchant **104** is associated with a merchant computing system **128**. The merchant computing system **128** is generally structured to process payments associated with purchases and returns from the merchant **104** (e.g., a payment associated with a purchase by the user **102** from the merchant **104**). The merchant computing system **128** includes a network interface **130** and a payment processing circuit **132**. The network interface **130** is structured to facilitate data communication with other computing systems (e.g., the user device **110**, the FI computing system **108**, etc.) via the network **112**. The network interface **130** includes hardware and program logic that facilitates connection of the merchant computing system **128** to the network **112** in a similar manner as described above with respect to the network interface **114** of the FI computing system **108**. The payment processing circuit **132** is structured to receive payment information (e.g., credit card information, checking account information, math-based currency information, etc.) from customers (e.g., the user **102**) and computing devices associated with customers (e.g., the user device **110**), and to forward the payment information to a payment processor (e.g., the FI computing system **108**, a credit card network computing system, a payment network computing system, etc.) as a payment request. Additionally, the payment processing circuit **132** is structured to receive approvals and denials from the payment processor associated with payment requests.

As described in further detail below with respect to FIG. 4, the FI computing system **106** is generally structured to

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authorize a transaction associated with the user **102** (e.g., a purchase by the customer from the merchant **104**, a transfer of funds, etc.) based at least in part on the location of the user device **110**. Accordingly, the user device **110** is structured to receive location information from various location devices and to transmit the location information to the FI computing system **108** via the network **112**. In some arrangements, the location devices include at least one locator beacon **134**. Each locator beacon **134** is structured to wirelessly transmit a unique identifier via a wireless transmitter (e.g., a Bluetooth transmitter). When the user device **110** is within a broadcast range of a given locator beacon **134**, the user device **110** can receive the locator beacon, and the location of the user device **110** can be determined (either locally by the user device **110** or remotely by the FI computing system **108**) by cross-referencing the received unique identifier with a location database. In other arrangements, the location device includes at least one wireless access point ("WAP") **136**. A WAP **136** may be, for example, a wireless router, a WAP associated with a business, or the like. The WAP **136** broadcasts a network identifier (e.g., an SSID), which is received by the user device **110** when the user device **110** is within a broadcast range of a given WAP **136**. If the user device **110** receives the network identifier, the location of the user device **110** can be determined (either locally by the user device **110** or remotely by the FI computing system **108**) by cross-referencing the received network identifier with a location database. In further arrangements, the location device includes at least one cellular transceiver **138**. Similar to the beacon **134** and the WAP **136**, the cellular transceiver **138** (e.g., a cell tower) transmits a unique cellular identifier. Accordingly, when the user device **110** is within broadcast range of the cellular transceiver **138**, the user device **110** is known to be within the vicinity of the cellular transceiver **138**. In further arrangements, the location devices include a combination of locator beacons **134**, WAPs **136**, and cellular transceivers **138**. Although the system **100** only shows locator beacons **134**, WAPs **136**, and cellular transceivers **138**, it should be understood that the user device **110** can receive location information from other sources, such as scanning a barcode or QR code displayed on a device having a known and fixed location, from GPS (or similar system) satellites, and the like.

The location information used by the FI computing system **106** may be received in the form of a digital fingerprint of the user device. The digital fingerprint is formed by a combination of device identifier information (e.g., a device serial number associated with the user device **110**) and location information associated with the user device **110**. Accordingly, the digital fingerprint changes depending on the location of the user device **110**. The location information may be formed by a received wireless signals (e.g., from any beacons **134**, WAPs **136**, or cellular transceivers **138** in broadcast range of the user device **110**), codes scanned from a designated location (e.g., a QR code scanned by the user device **110** from the screen of the computing device **202** or another known location), or the like. Since the digital fingerprint is specific to the user device **110**, the digital fingerprint is difficult for a fraudster attempting to transfer funds to spoof.

Referring to FIG. 2, a view of a user authentication system **200** is shown according to an example embodiment. The user authentication system **200** is similar to the system **100**. Accordingly, like numbering is maintained between FIG. 1 and FIG. 2 to designate like objects between the systems **100** and **200**. The primary difference between the system **100** and the system **200** is that the system **200** facilitates the authentication of the user **102** to a computing device **202** instead of

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facilitating the authentication of the user **102** as the originator of a transfer of funds (as done in the system **100**). The computing device **202** may be, for example, a work-access terminal, a personal computing device, a financial institution computing device, or the like. For example, the user **102** may be an employee attempting to gain access to a workstation (i.e., the computing device **202**) at the user's workplace.

Generally, to access the computing device **202**, the user **102** must authenticate himself as an authorized user of the computing device **202**. For example, if the user **102** is an employee and the computing device **202** is an employer work terminal, the user **102** must authenticate that the user is an employee who has access rights to the work terminal. This is traditionally achieved by the user **102** providing authentication information (e.g., username, password, token, biometric, etc.) to the computing device **202** which is either verified locally at the computing device **202** or remotely by a backend authentication system **204**. If the provided authentication information matches verified authentication information, the user **102** is provided access to the computing device **202**. However, authentication credentials can be easily spoofed by a person that is not authorized to access the computing device **202** (e.g., a hacker, a criminal, etc.). To add an additional or alternate authentication layer to the user authentication process, the system **200** additionally or alternatively relies on a digital fingerprint of a user device **110** known to be associated with the user **102**. The digital fingerprint is formed by a combination of device identifier information (e.g., a device serial number associated with the user device **110**) and location information associated with the user device **110**. Accordingly, the digital fingerprint changes depending on the location of the user device **110**. The location information may be formed by a received wireless signals (e.g., from any beacons **134**, WAPs **136**, or cellular transceivers **138** in broadcast range of the user device **110**), codes scanned from a designated location (e.g., a QR code scanned by the user device **110** from the screen of the computing device **202** or another known location), or the like. At the time of user authentication at the computing device **202**, the user device **110** provides a digital fingerprint at the same time which is verified as corresponding to the location of the computing device **202**. If any of the digital fingerprint or the user authentication information is not verified, the user **102** is not provided access to the computing device **202**. Since the digital fingerprint of the user device **110** is specific to the location of the computing device **202**, the digital fingerprint is difficult for an unauthorized user to spoof while attempting to gain unauthorized access to the computing device **202**. The authentication process is described in further detail below with respect to FIG. 5.

As noted above, the system **200** includes the backend authentication system **204**. The backend authentication system **204** includes a network interface **206**. The network interface **206** is structured to facilitate data communication with other computing systems (e.g., the user device **110**, the computing device **202**, etc.) via the network **112**. The network interface **206** includes hardware and program logic that facilitates connection of the backend authentication system **204** to the network **112**. For example, the network interface **206** may include a wireless network transceiver (e.g., a cellular modem, a Bluetooth transceiver, a WiFi transceiver, etc.) and/or a wired network transceiver (e.g., an Ethernet transceiver). In some arrangements, the network interface **206** includes the hardware and programming logic sufficient to support communication over multiple channels

of data communication (e.g., the Internet and an internal private network). Further, in some arrangements, the network interface **206** is structured to encrypt data sent over the network **112** and decrypt received encrypted data.

The backend authentication system **204** includes a processing circuit **208** having a processor **210** and memory **212**. The processor **210** may be implemented as a general-purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a digital signal processor (DSP), a group of processing components, or other suitable electronic processing components. The memory **212** includes one or more memory devices (e.g., RAM, NVRAM, ROM, Flash Memory, hard disk storage, etc.) that store data and/or computer code for facilitating the various processes described herein. Moreover, the memory **120** may be or include tangible, non-transient volatile memory or non-volatile memory.

The backend authentication system **204** includes an authentication circuit **214**. Although shown as separate circuits in FIG. 2, in some arrangements, the authentication circuit **214** is part of the processing circuit **208**. Other arrangements may include more or less circuits without departing from the spirit and scope of the present disclosure. Further, some arrangements may combine the activities of one circuit with another circuit to form a single circuit. Therefore, those of ordinary skill in the art will appreciate that the present arrangement is not meant to be limiting. The authentication circuit **214** is structured to authentication individuals attempting to access the computing device **202**.

The backend authentication system **204** includes an accounts database **216**. In some arrangements, the accounts database **216** is part of the memory **212**. The accounts database **216** is configured to hold, store, categorize, and otherwise serve as a repository for information associated with user accounts permitted to access the computing device **202**. For example, the accounts database **216** may store user identifiers, user passwords, user biometric information, user device information (e.g., the serial numbers of user devices associated with authorized users of the computing device **202**), verified user device digital fingerprint information, and the like. The accounts database **216** is accessed by the authentication circuit **214** during authentication of a user attempting to access the computing device **202**.

In an alternative arrangement, the functionality of the backend authentication system **204** is incorporated directly into the computing device **202**. In such an arrangement, the user authentication information and the digital fingerprint of the user device **110** are provided to the computing device **202** and verified locally at the computing device **202**.

Referring to FIG. 3, a table **300** showing information used to form a digital fingerprint of the user device **110** is shown. The digital fingerprint includes information that is gathered by the user device **110**. The digital fingerprint includes a device identifier **302** and a time **304** of the digital fingerprint. The device identifier **302** is a unique identifier that is associated with the user device **110**. The device identifier **302** may be, for example, an electronic serial number ("ESN"), an international mobile station equipment identity ("IMEI"), a mobile equipment identifier ("MEID"), a media access control address ("MAC address"), or the like. The time **304** may correspond to a device time maintained by the user device **110** or an externally maintained time. In some arrangements, the time **304** includes a date and a time.

The digital fingerprint includes a location snapshot **306** of the user device **110**. The location snapshot **306** is formed from various location information received at the user

device **110** at the time **304** associated with the digital fingerprint. The location snapshot **306** is formed from any combination of GPS satellite information **308**, WiFi SSID information **310**, locator beacon identifiers **312**, and other locator information (e.g., scanned QR code information **314**, scanned NFC tag identifiers **316**, etc.). The information listed in the table **300** is not intended to be limiting to the types of information that may be used to generate a digital fingerprint. For example, additional location information can be used to form the location snapshot **306**, such as GLONASS satellite information, cellular tower information (e.g., from cellular transceiver **138**), image information, and the like.

In some arrangements, the table **300** constitutes the digital fingerprint. In such arrangements, the user device **110** can transmit the table **300** to a receiving device (e.g., the FI computing system **108**, the backend authentication system **204**, the computing device **202**, etc.). In other arrangements, the table **300** is used by the user device **110** to generate a digital fingerprint. In such arrangements, the user device **110** transforms the information contained in the table **300** to create an alpha-numeric string that represents the information in the table **300** (e.g., by hashing the information contained in the table **300**). In both arrangements, the digital fingerprint may be encrypted by the user device **110** prior to transmission to the receiving device.

Referring to FIG. 4, a flow diagram of a method **400** of authenticating a financial transaction is shown according to an example embodiment. The method **400** is performed by the FI computing system **108** in the context of the system **100**. Specifically, the method **400** is performed by the payment approval circuit **124** of the FI computing system **108**. In some arrangements, the method **400** is performed during a transaction between the user **102** and the merchant **104**. In other arrangements, the method **400** is performed during a transaction in which the user **102** is providing payment to another entity (e.g., during a transfer of funds, during a purchase, during a donation, etc.). The other entity may be another financial institution.

The method **400** begins when a transaction request is received at **402**. The transaction request is received by the FI computing system **108** via the network **112**. In some arrangements, the transaction request relates to a transaction (e.g., a purchase) between the user **102** and the merchant **104**. In such arrangements, the FI computing system **108** receives the transaction request from the merchant computing system **128** or from a payment processing computing system (e.g., a point-of-sale system associated with the merchant **104**, a payment network computing system, etc.). In other arrangements, the transaction request relates to a transaction involving the transfer of funds out of an account associated with the user **102** to a destination (e.g., another account at another financial institution, an account associated with another individual or company, etc.). The transfer of funds may be associated with a wire transfer, a peer-to-peer payment, an ACH transfer, or the like. In such arrangements, the FI computing system **108** receives the transaction request from an initiating computing device (e.g., the user device **110**, a tablet computer, a laptop/desktop, a work computer, an ATM, a teller computing system, etc.) accessed by the user **102** or on behalf of the user **102**. The initiating computing device allows the user **102** (or an individual) to perform transactions associated with accounts held at the financial institution **106**. The transaction request includes information relating to the transaction, such as amount of money involved in the transaction, an identity of the user

102, an identity of the requestor (if the requestor is other than the user 102), an identity of the destination, and the like.

In some arrangements, the method 400—particularly steps 404 through 410—are only performed if a transaction condition is met. The condition may relate to the amount of money involved in the transaction being above a threshold amount or if the transaction is flagged as potentially fraudulent. In such arrangements, the information associated with the transaction received with the transaction request at 402 is compared against the threshold amount or for potential fraud indicators before proceeding to step 404. In further arrangements, the condition may relate to the transaction occurring at a designated period of time (e.g., a certain period of the week, a designated date, a designated time window, etc.), the transaction causing the number of transactions to exceed a designated amount of transactions during a time period (e.g., the transaction is the tenth transaction to occur in the same day), the type of transaction occurring (e.g., purchase vs. return), the transaction occurring during a user-indicated travel period, or the like. In other arrangements, all transactions associated with the user 102 or the account involved with the transaction are processed via the method 400. The description of the method 400 continues under the assumption that the transaction is processed via the method 400.

Location information of the user device is requested at 404. The FI computing system 108 transmits a request to the user device 110 associated with the user 102 that owns the account involved in the transaction. The user device 110 was previously bound or registered to the customer's account (e.g., by downloading a mobile banking application and completing the sign in process for the first time, by registering the user device 110 with the FI computing system 108, etc.). In some arrangements, the requested location information relates to a current digital fingerprint of the user device 110 (e.g., as described above with respect to FIG. 3). In other arrangements, the request causes the user 102 to be prompted via the user device 110 to enter location information (e.g., by scanning a QR code, by scanning an RFID tag, etc.). The location information may be static (e.g., a QR code that remains the same that is stored at a verified location of the user) or dynamic. For example, the financial institution 106 may request that the user log into an internet-banking website associated with the financial institution 106 where a dynamic transaction code (e.g., QR code) is presented for the user 102 to scan via the user device 110. The location information is used to verify that the user 102 is at an approved location during the transaction (e.g., a location where the user 102 can access the financial institution's website, a location where the user 102 has authorized transactions, such as the user's home or work, a location associated with the merchant 104, etc.). In some arrangements, if the user 102 did not initiate the transaction, the user 102 can respond to the inquiry from the FI computing system 108 to deny the transaction and to trigger a fraud alert or inquiry.

Location information is received from the user device at 406. The FI computing system 108 receives the location information from the user device 110 via the network 112. In some arrangements, the location information is encrypted. In such arrangements, the FI computing system 108 decrypts the encrypted location information. The FI computing system 108 determines whether the location information received at 406 corresponds to an approved location at 408. The FI computing system 108 compares the received location information with expected or verified location information to determine whether there is a match.

If the location information does not correspond to an approved location, the transaction is denied at 410. If the location information does not match the verified or expected location information, the transaction is denied. For example, if the expected location information relates to a known dynamic QR code, and the received location information includes a different code or no code, the location information does not correspond to an approved location. As another example, if the transaction is originating at a location of the merchant 104 (e.g., at a store), and the location information provided by the user device 110 corresponds with the user 102 being in a different location (e.g., at home, at work, in a different city, etc.), the location information does not correspond to an approved location. After the transaction is denied, the FI computing system 108 may send an alert to the user 102 to inform the user that a transaction associated with the user's account was attempted and denied.

If the location information corresponds to an approved location, the transaction is approved at 412. The FI computing system 108 transmits an approval message to the transaction requestor (e.g., to the merchant point of sale system, to the user 102 via the user device 110, etc.), and begins processing the transaction. In some arrangements, the FI computing system 108 initiates a transfer of funds (e.g., via ACH, via wire, etc.) in response to approving the transaction.

Referring to FIG. 5, a flow diagram of a method 500 of authenticating a user is shown according to an example embodiment. The method 500 is performed by the backend authentication system 204. Specifically, the method 500 is performed by the authentication circuit 214 of the backend authentication system 204. The method 500 is used to authenticate the user 102 at a computing device 202 (e.g., an employee workstation) based on user-provided authentication credentials and based on location information provided by a user device 110 associated with the user 102.

The method 500 begins with a login request is received at 502. The login request is initiated by the user 102 at the computing device 202. The backend authentication system 204 receives the login request from the computing device 202 via the network 112. The login request includes at least a user identifier associated with the user 102. For example, the login request may include a username, a user token, a user biometric, or the like. In some arrangements, the login request also includes at least one additional authentication factor, such as a password, a token, a biometric, a quasi-randomly generated number, or a combination thereof.

Location information is requested from the user device at 504. The backend authentication system 204 cross-references the accounts database 216 to verify the information provided with the login request as corresponding to known and verified information relating to the user 102. If the information provided does not match the known and verified information, the method 500 ends and the user 102 (or person attempting to impersonate the user 102) is not provided access to the computing device 202. However, the description of the method continues under the presumption that the provided information at 502 is valid. At this point, the backend authentication system 204 performs a location check on a user device 110 associated with the user 102 (e.g., the user's smartphone) to verify that the user device 110 is also in the location of the computing device 202. If the user device 110 is in a completely different location, there is a chance that the user 102 is also in the different location, and that the login attempt originates from a fraudster. Accordingly, the backend authentication system 204 transmits a request to the user device 110 associated with the user 102

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attempting to login to the computing device **202**. The user device **110** was previously bound or registered to the user **102** (e.g., during employee orientation, by binding the device during a prior login, etc.).

In some arrangements, the requested location information relates to a current digital fingerprint of the user device **110** (e.g., as described above with respect to FIG. **3**). In other arrangements, the request causes the user **102** to be prompted via the user device **110** to enter location information (e.g., by scanning a three-dimensional barcode such as a QR code, by scanning an RFID tag, etc.). The location information may be static (e.g., a QR code that remains the same that is stored at the location of the computing device **202**) or dynamic. For example, the computing device **202** display may present a dynamic transaction code (e.g., QR code) that is generated by and synchronized with the backend authentication system **204** that is presented for the user **102** to scan via the user device **110**. The location information is used to verify that the user **102** is at the location of the computing device **202**. In some arrangements, if the user **102** did not initiate the login request, the user **102** can respond to the inquiry from the backend authentication system **204** to deny that the user **102** originated the login request and to trigger a fraud alert or inquiry. The user's indication that the user **102** did not initiate the login request, the login request information may be used as a data point to improve the intelligence of a fraud detection system (e.g., by identifying the IP address of a computer responsible for the fraudulent login request, by blacklisting the IP address of the computer responsible for the fraudulent login request, etc.).

Location information is received at **506**. The backend authentication system **204** receives the location information from the user device **110** via the network **112**. In some arrangements, the location information is encrypted. In such arrangements, the backend authentication system **204** decrypts the encrypted location information. The backend authentication system **204** determines whether the location information corresponds to an approved location at **508**. The backend authentication system **204** compares the received location information with expected or verified location information to determine whether there is a match. For example, if the location information relates a digital fingerprint of the user device **110**, the backend authentication system **204** compares the received digital fingerprint with a known digital fingerprint (e.g., ensuring that the provided WiFi SSIDs, GPS signals, Bluetooth beacon identifiers, etc. match those of a previously registered and verified digital fingerprint associated with the user device **110**). As another example, if the location information relates to a scanned code, the backend authentication system **204** compares the received scanned code with the known code.

If the location information corresponds to an approved location, the user is authenticated at **510**. The backend authentication system **204** sends an authorization signal to the computing device **202**, and the computing device **202** provides the user **102** access to the computing device **202**. After access is provided, the method **500** ends.

If the location information does not correspond to an approved location, the method **500** continues down one of two branches at **512**. In the first branch of method **500**, the person attempting to access the computing device **202** is denied access to the computing device at **512**. In such arrangements, the backend authentication system **204** prevents the user **102** (or the person purporting to be the user **102**) from accessing the computing device **202**. The method **500** ends if access is denied to the user **102**.

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In the second branch, the method **500** does not immediately end. Rather, additional authentication information is requested at **512**. In such arrangements, the backend authentication system **204** requests additional authentication information from the user **102** (or the person purporting to be the user **102**) via the display of the computing device **202**. The additional authentication information may relate to, for example, a password associated with the user **102**, a biometric associated with the user **102**, the answer to a security question known by the user, or a combination thereof. This arrangement accounts for situations in which the user **102** leaves the user device **110** at a different location (e.g., if the user **102** leaves his smartphone at home but the user **102** is at the office). At **514**, the requested additional authentication information is received. The backend authentication system **204** receives the authentication information from the user **102** via the computing device. The backend authentication system **204** then authenticates the user based on the additional authentication information at **516**. If the provided information matches known and verified information, the user **102** is granted access to the computing device **202**, and the method **500** ends. If the provided information does not match known and verified information, the user **102** (or the person purporting to be the user **102**) is denied access to the computing device **202**, and the method **500** ends.

In an alternative arrangement, the computing device **202** is capable of self-authenticating the user in the same manner as described above with the backend authenticating system **204**. In such arrangements, the computing device **202** performs the steps of the method **500**.

The above-described authentication systems and methods provide for more secure transactions and more secure computer access systems. The systems and methods utilize location information related to a user device associated with a user involved with a transaction or login attempt. The location information may be packaged as a digital fingerprint, which can only be recreated by the specific user device associated with the user. Accordingly, the digital fingerprint is difficult—if not impossible—to spoof by a fraudster or by another device associated with the fraudster.

The embodiments described herein have been described with reference to drawings. The drawings illustrate certain details of specific embodiments that implement the systems, methods and programs described herein. However, describing the embodiments with drawings should not be construed as imposing on the disclosure any limitations that may be present in the drawings.

It should be understood that no claim element herein is to be construed under the provisions of 35 U.S.C. § 112(f), unless the element is expressly recited using the phrase “means for.”

As used herein, the term “circuit” may include hardware structured to execute the functions described herein. In some embodiments, each respective “circuit” may include machine-readable media for configuring the hardware to execute the functions described herein. The circuit may be embodied as one or more circuitry components including, but not limited to, processing circuitry, network interfaces, peripheral devices, input devices, output devices, sensors, etc. In some embodiments, a circuit may take the form of one or more analog circuits, electronic circuits (e.g., integrated circuits (IC), discrete circuits, system on a chip (SOCs) circuits, etc.), telecommunication circuits, hybrid circuits, and any other type of “circuit.” In this regard, the “circuit” may include any type of component for accomplishing or facilitating achievement of the operations described herein. For example, a circuit as described herein

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may include one or more transistors, logic gates (e.g., NAND, AND, NOR, OR, XOR, NOT, XNOR, etc.), resistors, multiplexers, registers, capacitors, inductors, diodes, wiring, and so on).

The “circuit” may also include one or more dedicated processors communicatively coupled to one or more dedicated memory or memory devices. In this regard, the one or more dedicated processors may execute instructions stored in the dedicated memory or may execute instructions otherwise accessible to the one or more dedicated processors. In some embodiments, the one or more dedicated processors may be embodied in various ways. The one or more dedicated processors may be constructed in a manner sufficient to perform at least the operations described herein. In some embodiments, the one or more dedicated processors may be shared by multiple circuits (e.g., circuit A and circuit B may comprise or otherwise share the same processor which, in some example embodiments, may execute instructions stored, or otherwise accessed, via different areas of memory). Alternatively or additionally, the one or more dedicated processors may be structured to perform or otherwise execute certain operations independent of one or more co-processors. In other example embodiments, two or more processors may be coupled via a bus to enable independent, parallel, pipelined, or multi-threaded instruction execution. Each processor may be implemented as one or more general-purpose processors, application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), digital signal processors (DSPs), or other suitable electronic data processing components structured to execute instructions provided by memory. The one or more dedicated processors may take the form of a single core processor, multi-core processor (e.g., a dual core processor, triple core processor, quad core processor, etc.), microprocessor, etc.

Any foregoing references to currency or funds are intended to include fiat currencies, non-fiat currencies (e.g., precious metals), and math-based currencies (often referred to as cryptocurrencies). Examples of math-based currencies include Bitcoin, Litecoin, Dogecoin, and the like.

It should be noted that although the diagrams herein may show a specific order and composition of method steps, it is understood that the order of these steps may differ from what is depicted. For example, two or more steps may be performed concurrently or with partial concurrence. Also, some method steps that are performed as discrete steps may be combined, steps being performed as a combined step may be separated into discrete steps, the sequence of certain processes may be reversed or otherwise varied, and the nature or number of discrete processes may be altered or varied. The order or sequence of any element or apparatus may be varied or substituted according to alternative embodiments. Accordingly, all such modifications are intended to be included within the scope of the present disclosure as defined in the appended claims.

The foregoing description of embodiments has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure to the precise form disclosed, and modifications and variations are possible in light of the above teachings or may be acquired from this disclosure. The embodiments were chosen and described in order to explain the principals of the disclosure and its practical application to enable one skilled in the art to utilize the various embodiments and with various modifications as are suited to the particular use contemplated. Other substitutions, modifications, changes and omissions may be made in the design, operating conditions and

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arrangement of the embodiments without departing from the scope of the present disclosure as expressed in the appended claims.

What is claimed is:

1. A method of authenticating a login request at a computing device, the method comprising:

receiving, by a computing device, the login request including a user identifier associated with a user;

transmitting, by the computing device, a request for location information of a user device associated with the user to indicate a first location of the user device;

causing, by the computing device, the user device to execute computer-executable code that is deployed to the user device to cause the user device to:

generate a first encoded value based on a unique identifier of the user device;

generate a second encoded value based on the first location of the user device; and

generate a location-based modifiable digital fingerprint comprising the first encoded value and the second encoded value, wherein the location-based modifiable digital fingerprint changes responsive to changes in location of the user device;

detecting, by the computing device, a change in location of the user device from the first location of the user device to a second location of the user device;

receiving, by the computing device, the location-based modifiable digital fingerprint from the user device based on the change in the location of the user device triggering the user device to change the location-based modifiable digital fingerprint;

verifying, by the computing device, the user device based on the change to the location-based modifiable digital fingerprint including a change to the second encoded value that corresponds to the second location of the user device; and

providing, by the computing device, the user access to the computing device.

2. The method of claim 1, wherein the login request further comprises a password associated with the user identifier.

3. The method of claim 2, further comprising verifying, by the computing device, that the user identifier and the password match a known user identifier and a known password.

4. The method of claim 1, wherein the first location of the user device is determined based on information provided by at least one of a wireless access point or a cellular transceiver in broadcast range of the user device.

5. The method of claim 1, wherein the first location of the user device is determined based on a code provided to the user device.

6. The method of claim 1, further comprising, in response to determining that the second encoded value corresponds to a third location of the user device, blacklisting, by the computing device, the user device.

7. The method of claim 6, further comprising blacklisting, by the computing device, the user device based on an IP address associated with the login request.

8. A system for authenticating a login request, the system comprising:

a computing device comprising a processing circuit configured to:

receive the login request including a user identifier associated with a user;

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transmit a request for location information of a user device associated with the user to indicate a first location of the user device;

cause the user device to execute computer-executable code that is deployed to the user device to cause the user device to:

- generate a first encoded value on a unique identifier of the user device;
- generate a second encoded value based on the first location of the user device; and
- generate a location-based modifiable digital fingerprint comprising the first encoded value and the second encoded value, wherein the location-based modifiable digital fingerprint changes responsive to a change in the location of the user device;

detect a change in the location of the user device from the first location of the user device to a second location of the user device;

receive the location-based modifiable digital fingerprint from the user device based on the change in the location of the user device triggering the user device to change the location-based modifiable digital fingerprint;

verify the user device based on the change to the location-based modifiable digital fingerprint including a change to the second encoded value that corresponds to the second location; and

provide the user access to the computing device.

9. The system of claim 8, wherein the login request further comprises a password associated with the user identifier.

10. The system of claim 9, wherein the processing circuit is further configured to verify that the user identifier and the password match a known user identifier and a known password.

11. The system of claim 8, wherein the first location of the user device is determined based on information provided by at least one of a wireless access point and a cellular transceiver in broadcast range of the user device.

12. The system of claim 8, wherein the first location of the user device is determined based on a code provided to the user device.

13. The system of claim 8, wherein the processing circuit is configured to, in response to determining that the second encoded value corresponds to a third location of the user device, blacklist the user device.

14. The system of claim 11, wherein the processing circuit is configured to blacklist the user device based on an IP address associated with the login request.

15. One or more non-transitory computer-readable media comprising instructions stored thereon, the instructions, when executed by a processor of a computing device,

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configured to cause the computing device to perform operations for authenticating a login request, the operations comprising:

- receiving the login request including a user identifier associated with a user;
- transmitting a request for location information of a user device associated with the user to indicate a first location of the user device;
- causing the user device to execute computer-executable code that is deployed to the user device to cause the user device to:
 - generate a first encoded value based on a unique identifier of the user device;
 - generate a second encoded value based on the first location of the user device; and
 - generate a location-based modifiable digital fingerprint comprising the first encoded value and the second encoded value, wherein the location-based modifiable digital fingerprint changes responsive to changes in location of the user device;
- detecting a change in location of the user device from the first location of the user device to a second location of the user device;
- receiving the location-based modifiable digital fingerprint from the user device based on the change in the location of the user device triggering the user device to change the location-based modifiable digital fingerprint;
- verifying the user device based on the change to the location-based modifiable digital fingerprint including a change to the second encoded value that corresponds to the second location of the user device; and
- providing the user access to the computing device.

16. The media of claim 15, wherein the login request further comprises a password associated with the user identifier.

17. The media of claim 16, the operations further comprising: verifying that the user identifier and the password match a known user identifier and a known password.

18. The media of claim 15, wherein the first location of the user device is determined based on information provided by at least one of a wireless access point and a cellular transceiver in broadcast range of the user device.

19. The media of claim 15, wherein the first location of the user device is determined based on a code provided to the user device.

20. The media of claim 15, the operations comprising: in response to determining that the second encoded value corresponds to a third location of the user device, blacklisting the user device.

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